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Private University established in Karnataka State by Act No. 84 of year 2010
Recognized by the University Grants Commission (UGC), New Delhi

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Bachelor of Computer

Applications Semester – V

Business Analytics

Experiment No.: 3

Title: Building a Dashboard

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Date: 04/09/2026

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Building a Dashboard

1. Introduction

Power BI is a business intelligence and data visualization tool used to transform raw data into meaningful and interactive reports. In this experiment, retail sales, customer, and product datasets were imported into Power BI and combined to create an interactive dashboard.

The dashboard focuses on sales performance, order quantity, product performance, customer demographics, annual income, product cost, product price, and sales trends. It provides management with a centralized view of retail business performance and supports data-driven decision-making.

2. Problem Statement

A retail company maintains sales, customer, and product information in separate datasets. Analysing these datasets individually makes it difficult to understand overall business performance.

Therefore, a centralized Power BI dashboard is developed to combine the datasets and analyse:

- Sales performance
- Order quantity
- Product performance
- Customer demographics
- Annual income
- Product price and cost
- Order trends over time
- Customer and product characteristics

The objective is to provide an interactive dashboard that enables management to easily understand business performance and identify important patterns.

3. Objectives

The main objectives of this experiment are:

1. To import retail business datasets into Power BI.
2. To clean and transform the datasets using Power Query.
3. To create relationships between Sales, Customer, and Product tables.
4. To analyse total sales and order quantity.
5. To study product-wise order performance.

6. To analyse customer distribution by gender.
7. To examine total children by gender.
8. To analyse annual income according to education level.
9. To compare product price with order quantity.
10. To develop an interactive and visually appealing Power BI dashboard.

4. Step 1: Data Collection

Three AdventureWorks datasets were used for the analysis.

Dataset	Description
AdventureWorks Sales Data 2022	Contains sales and order transaction information
AdventureWorks Customer Lookup	Contains customer demographic and income information
AdventureWorks Product Lookup	Contains product details, prices, costs and attributes

The Sales dataset contains 29,481 order lines covering the period from 1 January 2022 to 30 June 2022. It contains fields such as OrderDate, OrderNumber, ProductKey, CustomerKey, TerritoryKey and OrderQuantity.

The Customer dataset contains information about 18,154 customers, including Gender, MaritalStatus, AnnualIncome, TotalChildren, EducationLevel and Occupation.

The Product dataset contains 293 products and includes ProductName, ProductColor, ProductSize, ProductCost and ProductPrice.

Step 2: Data Import into Power BI

The datasets were imported into Power BI Desktop using the Get Data → Text/CSV option.

The following tables were imported:

- AdventureWorks Sales Data 2022
- AdventureWorks Customer Lookup
- AdventureWorks Product Lookup

The datasets were previewed before loading into Power BI. Column names and data types were also reviewed to ensure that the data was suitable for further processing.

Step 3: Data Cleaning and Transformation

Power Query Editor was used to clean and prepare the datasets.

The following operations were performed:

- Checked for missing values.
- Checked for duplicate records.

- Verified column names.
- Corrected inappropriate data types.
- Set OrderDate and StockDate as Date fields.
- Set OrderQuantity as a Whole Number.
- Set ProductPrice and ProductCost as Decimal Numbers.
- Verified ProductKey and CustomerKey across the datasets.
- Removed unnecessary columns.
- Applied the transformations and loaded the cleaned data into Power BI.

These steps helped ensure that the datasets were properly structured for analysis.

Step 4: Data Modeling

Relationships were created between the Sales table and the corresponding lookup tables.

Relationships created:

Sales Data 2022 → Product Lookup

ProductKey → ProductKey

Sales Data 2022 → Customer Lookup

CustomerKey → CustomerKey

Both relationships were created as one-to-many relationships, with the lookup tables filtering the Sales table.

Power BI's automatic Date Hierarchy was used with the OrderDate field to analyse order quantity by Year, Quarter, Month and Day.

Step 5: Measures and Aggregations

The dashboard uses Power BI's built-in aggregation functions to analyse the data.

The major aggregations used are:

- Sum of OrderQuantity
- Sum of Total Sales
- Sum of ProductPrice
- Sum of ProductCost
- Sum of AnnualIncome
- Sum of TotalChildren

The main order quantity aggregation is:

Sum of OrderQuantity = SUM('AdventureWorks Sales Data 2022'[OrderQuantity])

These aggregations are used across different visualizations to provide different perspectives of the retail data.

Step 6: Dashboard Development

A single-page Power BI dashboard was developed to analyse retail sales and customer performance.

The dashboard contains KPI cards, a line chart, scatter chart, pie chart, donut chart, bar chart, area chart, gauge visualization and interactive fields.

Total Order Quantity

A KPI card displays the total order quantity.

Total Order Quantity: 45K

The underlying dataset records 45,314 units sold during the reporting period.

This KPI provides an immediate overview of the total volume of products ordered.

Total Sales

The dashboard displays:

Total Sales: 9.19M

This KPI represents the overall sales value shown in the dashboard and provides a quick indication of business performance.

Product Price

The dashboard displays:

Sum of Product Price: 209.33K

This visualization provides an overview of the aggregated product price values in the dataset.

Order Quantity Trend Over Time

A line chart is used to display the Sum of OrderQuantity by Year, Quarter, Month and Day.

The visualization covers the period from January 2022 to June 2022. The chart shows considerable daily fluctuations in order quantity, with several noticeable peaks throughout the period.

The underlying analysis shows that monthly order quantity increased from 7,020 units in January to 8,260 units in June, indicating an overall increase in demand during the first half of 2022.

Order Quantity by Gender

A donut chart is used to analyse order quantity according to customer gender.

The dashboard shows approximately:

- Male – 23K (50.43%)
- Female – 22K (48.84%)

- NA – 0K (0.73%)

The underlying data records 22,851 units for male customers and 22,130 units for female customers.

The results indicate that order quantity is almost equally distributed between male and female customers.

Total Children by Gender

A pie chart represents the Sum of TotalChildren by Gender.

The dashboard displays approximately:

- Male – 17K (50.08%)
- Female – 16K (49.12%)
- NA – 270 (0.81%)

The visualization indicates that the number of children associated with male and female customer groups is relatively balanced.

Product Performance Analysis

A horizontal bar chart is used to analyse Sum of OrderQuantity by ProductName.

The updated dashboard displays products such as:

- Road Tire Tube
- Fender Set - Mountain
- Mountain Bottle Cage
- Road Bottle Cage
- Touring Tire Tube
- Sport-100 Helmet, Red
- Sport-100 Helmet, Blue
- ML Mountain Tire
- Sport-100 Helmet, Black
- LL Road Tire
- Touring Tire
- ML Road Tire

The chart makes it easier to identify products with comparatively high order quantities. This information can assist in inventory management, product planning and promotional decisions.

The original dataset analysis also indicates that several cycling accessories and related products contribute significantly to order volume.

Product Price and Order Quantity

A scatter chart compares:

- Sum of ProductPrice
- Sum of OrderQuantity

This visualization helps examine the relationship between product price and sales volume.

The analysis indicates a negative relationship between product price and order quantity, suggesting that lower-priced products tend to be purchased in greater quantities than premium-priced products.

This insight can help the business understand how pricing may influence product demand.

Annual Income by Education Level

An area chart displays Sum of AnnualIncome by EducationLevel.

The dashboard includes the following education categories:

- Bachelor's
- Partial College
- Graduate Degree
- High School
- Partial High School

The Bachelor's category has the highest aggregated annual income in the displayed dashboard, followed by Partial College, Graduate Degree, High School and Partial High School.

This visualization helps understand the income distribution of customers according to their educational background.

Annual Income, Product Cost and Order Quantity

A gauge-style visualization is used to display the combined measures of:

- Annual Income
- Product Cost
- Order Quantity

The visualization provides a high-level comparison of these measures and allows users to understand their relative scale within the dataset.

Customer and Product Analysis

The dashboard includes additional interactive fields that allow the data to be explored in greater detail.

Customer-related fields:

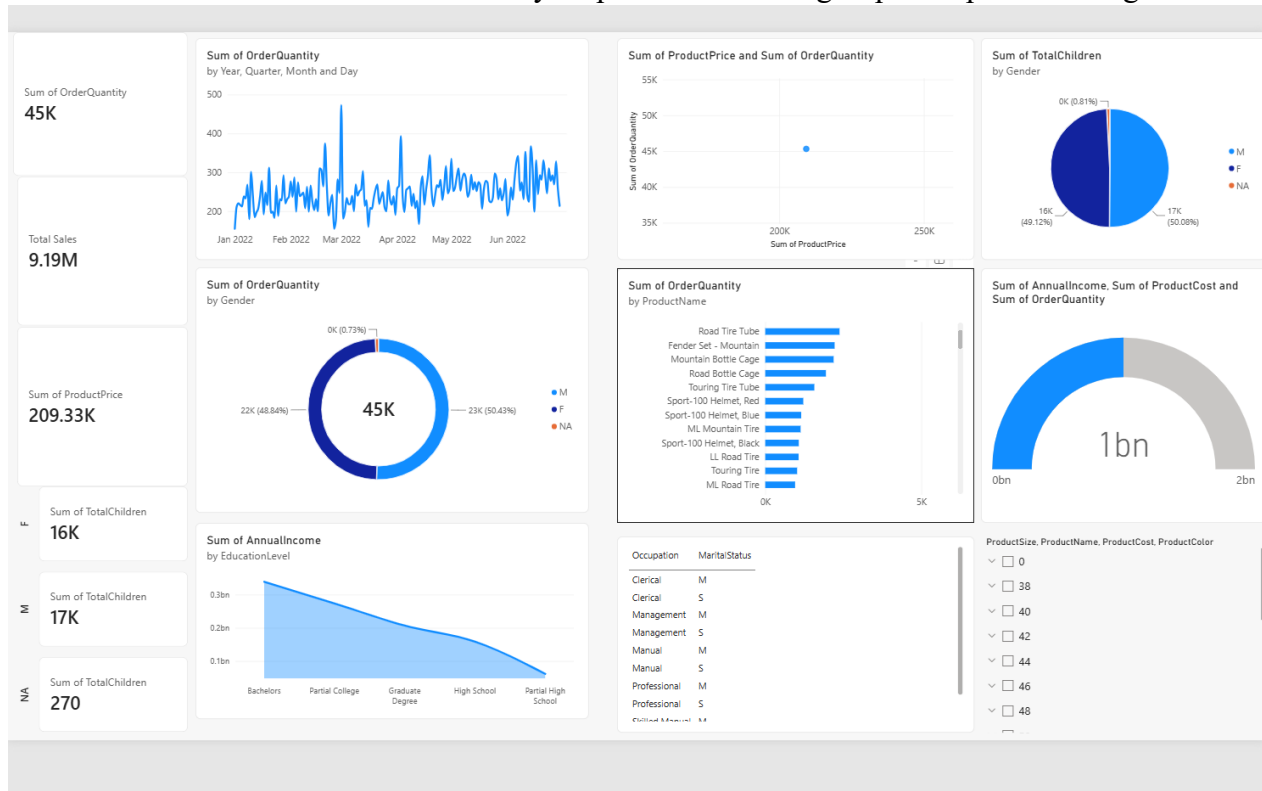
- Occupation

- Marital Status

Product-related fields:

- Product Size
- Product Name
- Product Cost
- Product Color

These fields allow users to filter and analyse specific customer groups and product categories.



Dashboard Formatting

The dashboard was designed as a single-page report with a clear and organized layout.

The following formatting techniques were applied:

- KPI cards were positioned for quick access to important values.
- Charts were arranged in a structured layout.
- A consistent visual theme was maintained.
- Product performance was represented using a horizontal bar chart.
- Time-based performance was represented using a line chart.
- Customer gender was represented using donut and pie charts.
- Product price and order quantity were compared using a scatter chart.
- Annual income was represented using an area chart.

- Interactive fields were included for customer and product analysis.

The dashboard therefore provides both a high-level overview and the ability to perform detailed analysis.

Key Findings and Insights

The major findings from the dashboard are:

1. The total order quantity is approximately 45K units.
2. The dashboard reports 9.19M in total sales.
3. The aggregated product price is approximately 209.33K.
4. Order quantity varies considerably from day to day.
5. Monthly order quantity shows an overall increase from January to June 2022.
6. Male and female customers contribute almost equally to order quantity.
7. Male customers contribute slightly more order quantity than female customers.
8. Total children are also distributed almost equally between male and female categories.
9. Several cycling products and accessories show high order quantities.
10. Lower-priced products generally tend to have higher order volumes.
11. Customers in the Bachelor's education category contribute the highest aggregated annual income in the displayed chart.
12. Product and customer filters allow more detailed analysis of the business data.

Final Dashboard

The final dashboard combines the Sales, Customer and Product datasets into a single interactive Power BI report.

The dashboard provides a centralized view of:

- Total Sales
- Total Order Quantity
- Product Price
- Product Performance
- Order Trends
- Customer Gender
- Total Children
- Annual Income
- Product Cost
- Education Level
- Occupation
- Marital Status

- Product Attributes

The completed dashboard transforms multiple datasets into a single visual reporting system that can be used to understand retail demand and customer behaviour.

Conclusion

The experiment successfully demonstrated the use of Power BI for retail sales and customer performance analysis. The AdventureWorks Sales, Customer and Product datasets were imported, cleaned and transformed using Power Query and connected through ProductKey and CustomerKey relationships.

An interactive dashboard was developed using KPI cards, line charts, bar charts, scatter charts, pie charts, donut charts, area charts and gauge visualizations.

The dashboard provides important insights into sales performance, order quantity, product demand, customer demographics, annual income and pricing patterns. With approximately 45K total order quantity and 9.19M in total sales, the dashboard provides management with a clear overview of retail performance.

Overall, the project demonstrates how Power BI can convert raw business data into meaningful visual insights and support data-driven decision-making in a retail environment.

Learning Outcomes

Through this experiment, I learned:

- Data Collection and Integration – how to work with multiple related business datasets.
- Power BI Data Import – how to import CSV datasets into Power BI.
- Data Cleaning and Transformation – how to use Power Query to clean data and correct data types.
- Data Modeling – how to create relationships between Sales, Customer and Product tables.
- Data Aggregation – how to use aggregation functions such as Sum of OrderQuantity, ProductPrice, ProductCost and AnnualIncome.
- Data Visualization – how to create KPI cards, bar charts, line charts, scatter charts, pie charts, donut charts, area charts and gauge visualizations.
- Customer Analysis – how to analyse customers based on gender, education, occupation, marital status and total children.
- Product Analysis – how to identify products with higher order quantities and analyse product pricing.
- Time-Based Analysis – how to analyse order trends across different dates and months.
- Business Intelligence – how to convert raw datasets into meaningful business insights.
- Data-Driven Decision Making – how an interactive Power BI dashboard can support business planning and management decisions.

Github Link:- <https://github.com/rzwaaahn/BA->